COURSE INFORMATION

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School/Faculty:	**Faculty of Artificial Intelligence**		
Program name:	**Bachelor in Artificial Intelligence with Honours**		
Course code:	**SAIA 1013**	**Academic Session/Semester:**	**3, 2024/25**
Course name:	**Responsible Artificial Intelligence and Ethics**	**Pre/co requisite (course name and code, if applicable):**	**Nil**
Credit hours:	**3**		

Course synopsis	This course addresses the ethical considerations and responsibilities that must be taken into account concerning artificial intelligence. With appropriate ethics and responsibility, the objective of this course is to equip students with the knowledge and guided skill set necessary to develop AI for the benefit of society and technology consumers. In addition, the course underlines the significance of comprehending the risk, impacts and threats posed by new developments in AI technology, as well as the application of present regulations. Ethical issues in AI, AI governance, AI system transformation, ethical risk assessment and management, ethical matrices, along with societal and educational AI ethics will be among the subjects covered.
Course coordinator (if applicable)	Dr. Nurazean Maarop

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr Nurazean Maarop	MJIIT 06.06.01	0169853093	nurazean.kl@utm.my
	Dr Ganthan Narayana Samy	MJIIT: 07.35.01	03-26154983	hazilah.kl@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	**CLO***	**PLO **(MQF Cluster Code)**	***Taxonomies and ****generic skills**	**T&L methods**	*****Assessment methods**
CLO1	Recommend resolutions to the ethical issues of Artificial Intelligence that are raised by the development, deployment and use of AI	PLO2 (C2)	C5 S	Lecture Think pair share Case Study	F T
CLO2	Explain methodologies and key approaches to AI ethics with a particular emphasis on virtue ethics to support and enhance human well-being in a sustainable environment	PLO5 (C3C)	A4 SI	Peer Assessment Reflection	Pr Asg
CLO3	Practise the appropriate ethical risk assessment and management using relevant framework according to the relevant ethics and professional code of practice	PLO11 (C5)	A5 A	Brainstorming Case Based Learning	CADR

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
*Up to 5 CLO
Refer ***Taxonomies of Learning and ****UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement
*****T – Test; Q – Quiz; HW – Homework; CADR - Case Analysis and Discussion Report; Asg – Assignment; PR – Project; Pr – Presentation; F – Final Exam etc.

****MQF Cluster Code**
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	**Type**	**Implementation**
1	Comparative Learning Strategy	In Assignments, students are divided into groups, with each group applying a different method or tool to solve the same problem or complete the same task. After completing the work, all groups present their methodologies and outcomes. This is followed by a discussion or analysis phase, where students compare and contrast the effectiveness, efficiency, and other aspects of each approach
2	Self-Directed Learning	Online tutorials, forums, and documentation to learn AI ethical concepts at their own pace.
3	Active Learning	Pedagogical approach that requires students to actively participate in their learning by participating in conversations, problem solving, case studies, role plays, and other interactive approaches.
4	Case-Based Learning	Students collaborate in groups to complete case studies, which are narratives that include one or more characters and/or settings. A disciplinary issue or difficulties are presented in the cases, and students are tasked with developing solutions to these problems under the direction of the lecturer.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
Communication Skills
Ethics and Professionalism

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	**Course Content Outline and Subtopics**	**CLO***	**Learning and Teaching Activities**										**TOTAL SLT**
			Face-to-Face (F2F)								**Non F2F Independent Learning**		
			Physical				**Online (Synchronous)**				**Online (Asynchronous)**	**Others**	
			L	**T**	**P**	**O**	**L**	**T**	**P**	**O**			

Week 1	**Introduction to Responsible AI and AI Ethics** (Understand the concept of Responsible AI and its significance in modern AI practices) - Ethics in Digital Era - The ART of AI (Accountability, Responsibility, Transparency) - Three Level of AI Transparency - Bias in AI - Fairness and Transparency - Privacy and Surveilance (Activities: Lecture, Discussion, Case Studies, Short Quiz)	CLO1	6								1		7
Week 2	**Theoretical Concept of Ethics and their Application to AI** - Major ethical theories (e.g., Utilitarianism, Deontology, Virtue Ethics). - Integration of Ethical Theories in AI - Ethical theory in Decision Making (Activities: Lecture, Group Discussion) - Human Vulnerabilities - Privacy & Data Protection	CLO1	6								1		7
Week 3	**Islamic Values in AI Ethics** - Related Policy, Legal, & Regulator - Conceptualizing Resolution of AI Ecosystems for Human Flourishing	CLO2	6								1		7
Week 4	**Ethics Centric in AI Design** - Principle of Designing AI Systems Priotizing Human Well-Being and Ethics - Shaping AI Ecosystems Resilience (Social and Development Perspective) - Modeling, Reasoning, Alignment for AI System Transformation	CLO1 CLO2	6								1		7
Week 5	**Risk Assessment for AI Ethics** - Fundamental Ethical Risk Assessment and Management of AI Project - The Ethical Matrix of AI Systems and Data Science	CLO3	6								1		7
Week 6	**Governance for Responsible AI** - Ethical Framework Associated with AI - Sustainable AI Ethics Framework & Methodology - Principle of Global Community Responsibility - Principle of Societal Responsibility - Relevant Standard and Roadmap for AI Ethics	CLO2 CLO3	6								1		7
Week 7	**Continous Improvement** - The Ethics of Artificial Intelligence in Education Practices - The Near Future of AI: Special Issue	CLO2 CLO3	3								1		4
	Case Discussion & Presentation	CLO2 CLO3				3					3	3	9
Week 8	Final Examination	CLO1											
											SUB-TOTAL SLT :		**55**

	Continuous Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Assignment	20	10		7		17
2	Presentation	10	6		7		13
3	Case Analysis and Discussion	25	5		12		17
4	Test	15	2		5		7
5							
					SUB-TOTAL SLT :		**54**

	Summative Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	30	3		8		11
2							0
3							0
4							0
5							0
					SUB-TOTAL SLT :		**11**
					SLT for Assessment:		**65**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	56.67
B	% SLT for Online & Independent Learning Component :	43.33
C	%SLT for Online Component:	40.83
D	% SLT for All Practical Component:	0.00
D1	% SLT for F2F Physical Practical Component:	0.00
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

References (include required and further readings, and should be the most current)

1. Stahl, B. C. (2021). Artificial intelligence for a better future: an ecosystem perspective on the ethics of AI and emerging digital technologies (p. 124). Springer Nature.
2. Dignum, V. (2019). Responsible artificial intelligence: how to develop and use AI in a responsible way (Vol. 2156). Cham: Springer.
3. Liao, S. M. (Ed.). (2020). Ethics of artificial intelligence. Oxford University Press.
4. Tzimas, T. (2021). Legal and Ethical Challenges of Artificial Intelligence from an International Law Perspective (Vol. 46). Springer Nature.
5. Thompson, S. J. (Ed.). (2021). Machine Law, Ethics, and Morality in the Age of Artificial Intelligence. IGI Global.
6. Beranger, J. (Ed.). (2021). Societal responsibility of artificial intelligence: Towards an ethical and eco-responsible AI. John Wiley & Sons.
7. Holmes, W., & Porayska-Pomsta, K. (Eds.). (2022). The Ethics of Artificial Intelligence in education: Practices, challenges, and debates. Taylor & Francis.

Other additional information (if applicable)

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

Disclaimer:

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While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours
Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:		**Certified by:**
Name:	Dr. Nurazean Binti Maarop	**Name:**
Signature:	Nurazean	**Signature:**
Date:	December 2023	**Date:**